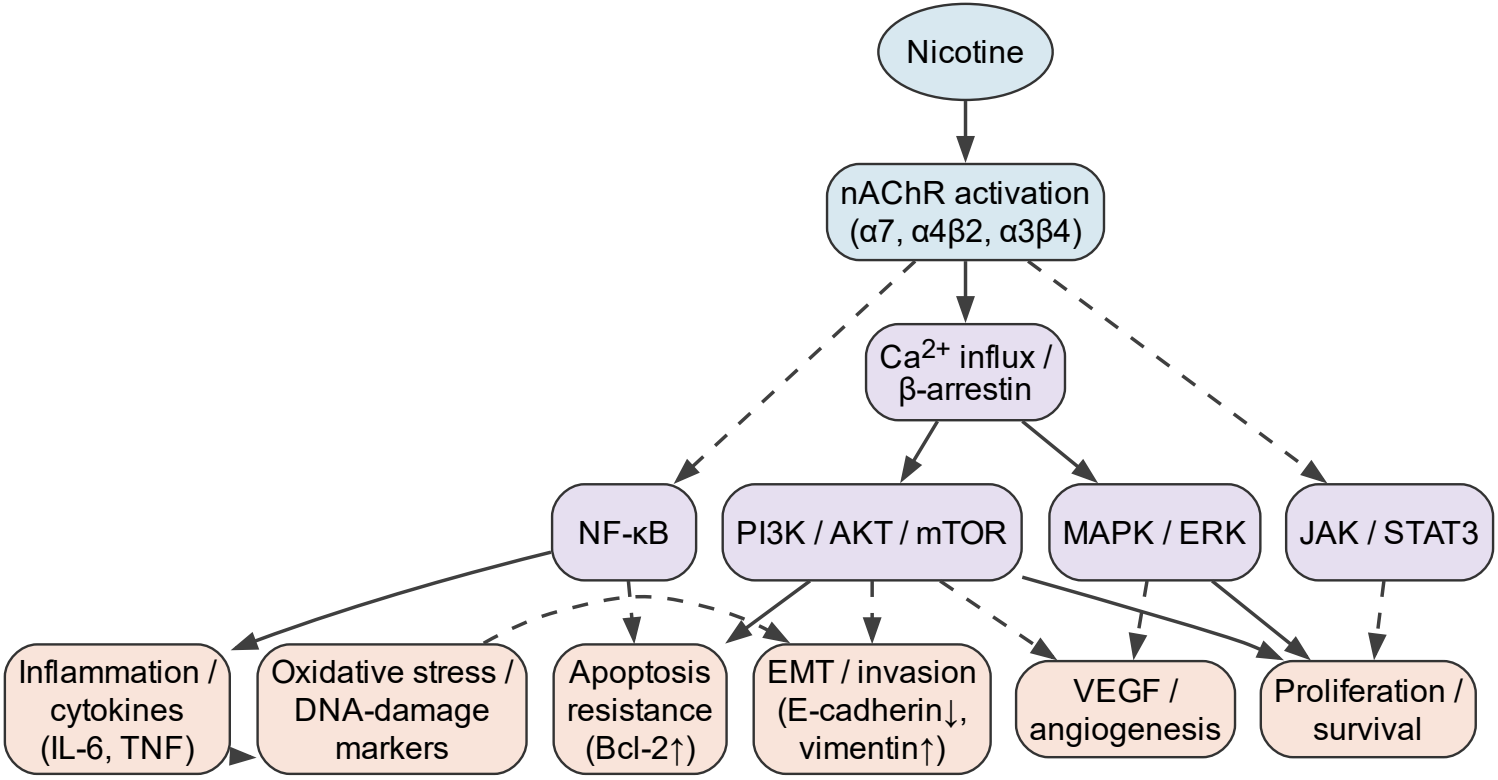
Nicotine
nAChR activation
(α7, α4β2, α3β4)
Ca2+ influx /
β-arrestin
NF-κB
PI3K / AKT / mTOR
MAPK / ERK
JAK / STAT3
Inflammation /
cytokines
(IL-6, TNF)
Oxidative stress /
DNA-damage
markers
Apoptosis
resistance
(Bcl-2↑)
EMT / invasion
(E-cadherin↓,
vimentin↑)
VEGF /
angiogenesis
Proliferation /
survival